FREE DEFENSIVE SECURITY GUIDE

Blue Team & SOC Analyst Roadmap 2026

From logs to detection, investigation, response, and Purple Team validation.

A practical, tool-aware roadmap for people starting defensive cybersecurity — without requiring an expensive lab or a university program.



Community Learning Project • Updated October 2026

What Blue Team work actually is

Defensive security is not “watching alerts.” It is building reliable evidence, judgment, and response.

THE MISSION

SEE

Collect the right telemetry from endpoints, identity, network, cloud, applications, and security controls.

DECIDE

Turn signals into hypotheses. Validate what happened, who was affected, and how confident you are.

ACT

Escalate, contain, preserve evidence, eradicate, recover, and communicate without making the incident worse.

IMPROVE

Write detections, tune noisy rules, close visibility gaps, update playbooks, and retest against realistic behavior.

Core mindset

Logs are evidence. Alerts are hypotheses. Your job is to investigate until the story makes sense.

How to use this roadmap

Do not memorize a tool list. Build one investigation skill at a time, then prove it with evidence.

LEARNING LOOP

1 LEARN

Understand the protocol, operating system, identity flow, or attack behavior.

2 INSTRUMENT

Generate or collect telemetry that would reveal the behavior.

3 INVESTIGATE

Search, pivot, correlate, build a timeline, and test competing explanations.

4 EXPLAIN

Write what happened in plain language with evidence and confidence.

5 IMPROVE

Create or tune a detection, playbook, query, or hardening action.

6 RETEST

Reproduce the behavior and verify that the defense now works.

Rule of thumb: every study session should answer one question better than you could yesterday.

The 12–18 month progression

Move in layers. Foundations first; specialization after you can reliably investigate evidence.

ROADMAP

01	02	03	04	05	06	07	08
FOUNDATIONS	**LOGS**	**SIEM**	**NETWORK**	**DETECTION**	**IR / DFIR**	**HUNTING**	**PURPLE**
Networking • Linux • Windows • DNS • HTTP/TLS	Windows events • Sysmon • auth • app logs	Search • fields • pivots • dashboards • alert triage	Wireshark • Zeek • Suricata • PCAP reasoning	Sigma • ATT&CK • baselining • tuning	scoping • timelines • artifacts • containment	hypothesis-driven hunts • intel • anomalies	simulate → observe → detect → improve → retest
0–2 mo	2–4 mo	3–6 mo	5–8 mo	7–11 mo	9–13 mo	11–15 mo	12–18 mo

Stage 1 — Foundations defenders actually use

A SOC analyst who understands normal systems behavior can recognize abnormal behavior much faster.

FOUNDATIONS

NETWORKING

TCP/UDP • subnetting • routing • NAT • DNS • DHCP • ARP • HTTP/HTTPS • TLS • SMB • common ports

Practice: packet captures, DNS lookups, connection states, client/server flows.

OPERATING SYSTEMS

Windows Event Viewer • services • registry • scheduled tasks • PowerShell • Linux processes • systemd • journald • permissions

Practice: explain what “normal” looks like before hunting malicious behavior.

IDENTITY

Active Directory • local vs domain accounts • Kerberos basics • NTLM • MFA • SSO • tokens • service accounts

Practice: trace an authentication from user to resource.

SCRIPTING

Python basics • PowerShell • Bash • regex • JSON • CSV • APIs

Goal: automate repetitive analysis, not become a software engineer first.

Telemetry before tooling

You cannot detect what you never collected. Learn what each data source can and cannot prove.

VISIBILITY

ENDPOINT

Windows Event Logs • Sysmon • EDR • process • file • registry • PowerShell

IDENTITY

AD/Entra sign-ins • Kerberos/NTLM • MFA • account changes • privilege use

NETWORK

DNS • flow • Zeek metadata • Suricata alerts • proxy • firewall • packet capture

CLOUD / APPS

Cloud audit logs • SaaS auth • web/app logs • database audit • API activity

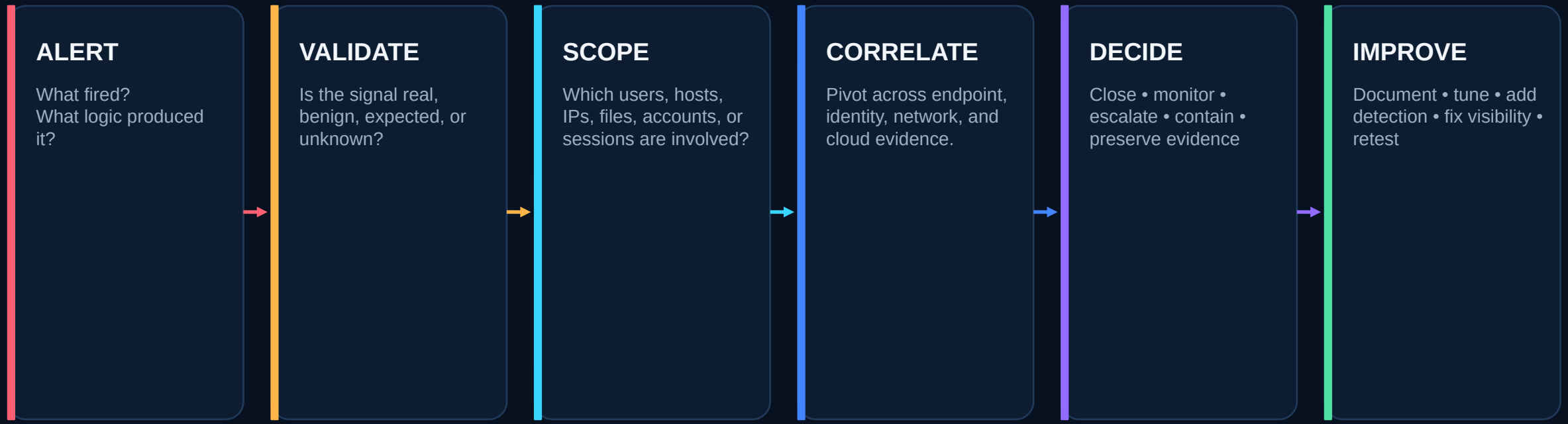


2026 note: Sysmon is available as a built-in optional feature for Windows 11, in addition to the traditional Sysinternals package.

The SOC alert lifecycle

An alert is the beginning of an investigation—not the conclusion.

WORKFLOW



A good Tier 1 analyst does not “solve every incident.” A good analyst gathers enough reliable evidence to make the next decision safe and efficient.

Modern defensive toolkit — organized by question

Tools are useful only when you know what evidence you are trying to obtain.

TOOL STACK

What happened on the endpoint?

Windows Event Logs • Sysmon • Velociraptor • osquery • Sysinternals

What crossed the network?

Wireshark • Zeek • Suricata • tcpdump

Where do I search at scale?

Wazuh • Security Onion • Elastic/OpenSearch-style SIEM workflows

How do I express detections?

Sigma • YARA • Suricata rules • native SIEM queries

How do I enrich evidence?

CyberChef • VirusTotal/public reputation • WHOIS/DNS • threat intel feeds

How do I validate defenses?

Atomic Red Team • safe lab simulations • known PCAP/log datasets

Do not install everything at once. Learn one tool deeply enough to answer one class of questions, then expand.

Investigation methodology — stay on one evidence thread

A repeatable method prevents “tool hopping” and makes your conclusions defensible.

DECISION FLOW

1 What triggered?

Read the alert logic. Identify the entity, time window, and expected behavior.

2 Can I validate the event?

Find the raw log / packet / process / authentication event behind the alert.

3 Is it normal here?

Compare with baseline, user role, parent process, destination, frequency, and business context.

4 What else is connected?

Pivot by user, host, IP, hash, process tree, session, time, and identity changes.

5 What is the impact?

Decide whether evidence supports benign activity, suspicious behavior, or confirmed malicious activity.

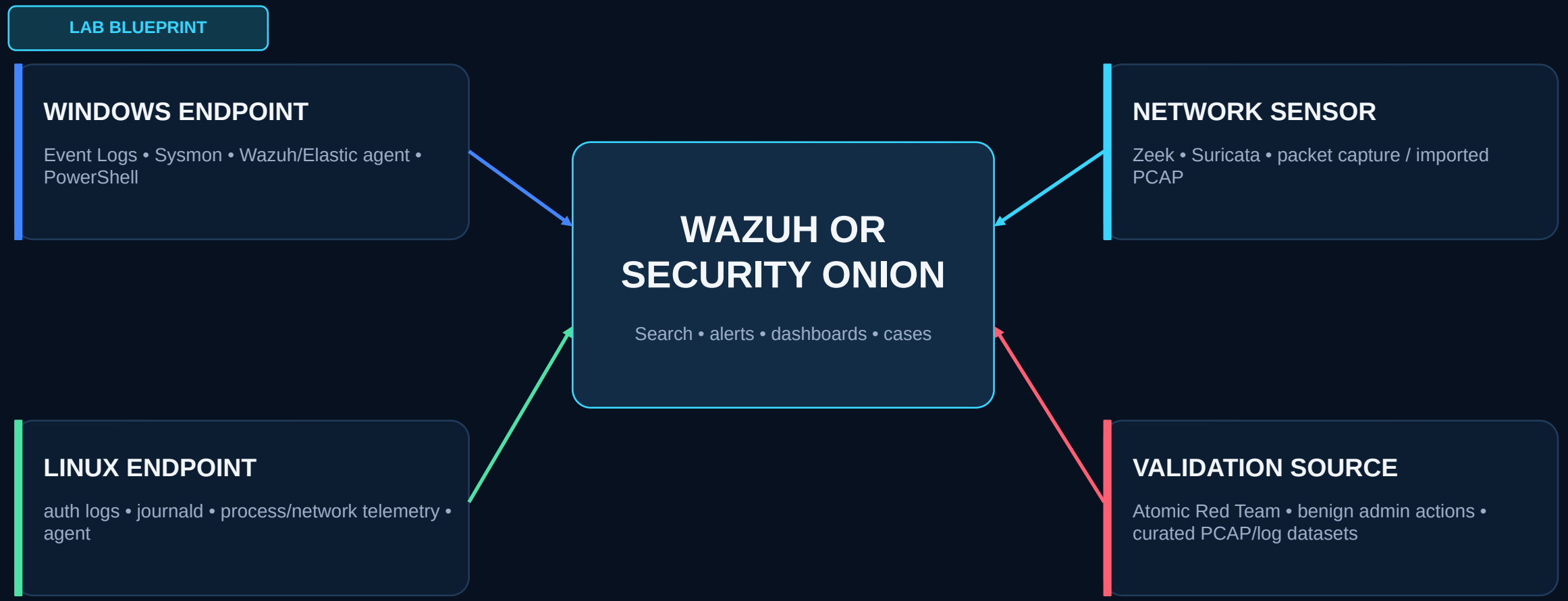
6 What is the safest next action?

Close, monitor, escalate, contain, preserve, or request additional telemetry.

Evidence hierarchy: raw telemetry > correlated context > tool verdict > intuition.

A lightweight lab that teaches real SOC skills

Start small. One SIEM, one Windows endpoint, one Linux endpoint, and reproducible test data are enough.



Budget rule: if setup consumes the study session, simplify the lab.

Detection engineering — from behavior to reliable signal

A detection should explain what behavior matters, what telemetry is required, and why the signal deserves attention.

DETECTION

BEHAVIOR

Choose a meaningful adversary or abuse behavior.

1

TELEMETRY

Identify the logs and fields that can observe it.

2

LOGIC

Write a query / Sigma rule / analytic that finds the pattern.

3

TEST

Generate known activity and confirm the signal appears.

4

TUNE

Reduce false positives without hiding the behavior.

5

MITRE ATT&CK v18 moved toward Detection Strategies and Analytics: think in behaviors + evidence requirements, not just “one rule per technique.”

Sigma remains a portable, YAML-based format for expressing log detections across SIEM ecosystems.

Incident response + DFIR — investigate without destroying the story

Containment matters, but evidence, scope, communication, and recovery matter too.

IR / DFIR

DETECT

Confirm the incident and establish confidence.

RESPOND

Scope, prioritize, contain, coordinate, and preserve what matters.

RECOVER

Restore safely, monitor for recurrence, and validate normal operations.

Evidence disciplines worth learning

- Timeline thinking: what happened before, during, and after the alert.
- Volatile vs persistent evidence: know what disappears when a host changes state.
- Chain of custody and documentation for cases that may become legal, HR, or compliance matters.
- Endpoint triage: processes, autoruns, services, tasks, registry, browser/artifact history, network connections.
- Network forensics: DNS, sessions, JA4/TLS metadata where available, HTTP, SMB, packet evidence.
- Recovery validation: the incident is not over just because the process was killed.

NIST SP 800-61 Rev. 3 (2025) aligns incident response with CSF 2.0 and emphasizes Detect → Respond → Recover with continuous improvement.

Threat hunting + Purple Team validation

Hunting asks a question. Purple teaming tests whether your defenses can answer it.

HUNT → TEST → IMPROVE

THREAT HUNTING

Start with a hypothesis, not a dashboard.

Example pattern:
“Could a compromised user execute unusual remote admin activity without generating a useful signal?”

Hunt across authentication, process, network, and remote-management telemetry.

PURPLE VALIDATION

Reproduce a safe, authorized behavior in the lab.

Use Atomic Red Team or controlled manual actions. Observe what telemetry appears, which detections fire, and what the analyst can actually conclude.

IMPROVEMENT LOOP

Add missing telemetry. Write or tune detections. Update playbooks. Improve enrichment. Re-run the same test.

Success is not “we generated an alert.”
Success is “an analyst can detect, investigate, and respond correctly.”

Free and genuinely useful places to learn

Prefer resources that make you inspect evidence, explain conclusions, or build detections.

HANDS-ON LEARNING

MITRE ATT&CK

Behavior mapping, Detection Strategies, Analytics, technique context

attack.mitre.org

Sigma HQ

Portable detection rules, rule structure, log-source thinking

sigmahq.io

Security Onion Docs

Free defender platform; Zeek, Suricata, endpoint visibility, hunt/cases

docs.securityonion.net

Wazuh Docs

Free/open SIEM + XDR platform; endpoint telemetry and threat hunting

documentation.wazuh.com

LetsDefend — free content

SOC Fundamentals, Windows Fundamentals, SIEM 101 + selected free practice

letsdefend.io

CyberDefenders

Blue-team CTF challenges covering network forensics, malware, cloud, logs

cyberdefenders.org

Malware-Traffic-Analysis.net

Current PCAP exercises with realistic malware/network traffic

www.malware-traffic-analysis.net

CISA / NIST

Incident-response guidance, no-cost training resources, playbooks, standards

www.cisa.gov/ | https://csrc.nist.gov

A practical 90-day starter sprint

Designed for people with limited time: learn, investigate, document, repeat.

90 DAYS

DAYS 1–30

FOUNDATIONS + LOGS

- Networking refresh
- Windows Event Viewer
- Sysmon / process trees
- Linux auth + journald
- Wireshark basics
- Write 4 mini incident notes

DAYS 31–60

SIEM + INVESTIGATION

- Deploy Wazuh or use a hosted/free lab
- Onboard Windows + Linux telemetry
- Practice searches and pivots
- Investigate 6 alerts / datasets
- Start MITRE ATT&CK mapping
- Build one timeline from evidence

DAYS 61–90

DETECTION + PORTFOLIO

- Write 3 Sigma rules
- Run safe validation tests
- Analyze 2 PCAP exercises
- Tune one noisy detection
- Create one incident report
- Publish sanitized findings to GitHub

After 90 days you should have evidence of skill: queries, notes, timelines, detections, PCAP analysis, and a small lab—not just completed videos.

Build proof, not just badges

A hiring manager cannot see what you learned in your head. Make the work visible.

PORTFOLIO

INVESTIGATION WRITE-UP

Alert summary • evidence • timeline • pivots • conclusion • confidence

DETECTION PROJECT

Hypothesis • required telemetry • Sigma/query • test data • false-positive notes

PCAP CASE

Traffic summary • DNS/HTTP/TLS observations • IOCs • timeline • ATT&CK mapping

SOC LAB

Architecture diagram • log sources • screenshots • sample searches • lessons learned

INCIDENT REPORT

Executive summary • technical findings • scope • impact • actions • recommendations

PURPLE TEST

Behavior simulated • expected telemetry • observed telemetry • detection gap • retest result

Use GitHub for the permanent technical record. Use LinkedIn to explain what you learned and invite professional feedback.

Honesty beats inflated titles: “Here is the evidence I analyzed, how I reasoned through it, what I got wrong, and what I improved.”

The defender’s operating principle

Understand normal. Collect evidence. Investigate deliberately. Improve the system.

See → Decide → Act → Improve → Retest

Primary references used for this 2026 edition

- MITRE ATT&CK — Detection Strategies / Analytics
- Sigma HQ — detection format and rule docs
- Microsoft Sysmon documentation
- Security Onion documentation
- Wazuh documentation
- NIST SP 800-61 Rev. 3 — Incident Response
- Zeek / Suricata / Velociraptor / osquery official docs

PROJECT ACKNOWLEDGMENT

This educational roadmap was created by Sebastian Romero with the assistance of ChatGPT by OpenAI.

Built to help beginners find a structured path into defensive cybersecurity and share practical knowledge freely with the community.

Use it. Question it. Improve it. Share what you learn.